

CROP YIELD PREDICTION USING MACHINE LEARNING TECHNIQUES: A CASE STUDY IN CAUVERY DELTA ZONE, TAMIL NADU, INDIA

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ABSTRACT:

Agriculture plays a vital role in supporting livelihoods and economies. Farmers face challenges like water scarcity, uncertain prices, weather fluctuations, and unpredictable crop yields. Predicting crop yield, particularly for crops like paddy, is complex due to various factors. While traditional statistical methods have been used, their accuracy is limited. This study explores machine learning techniques like random forest algorithm (RF), linear regression algorithm (LR), support vector machine (SVM) for accurate paddy yield

prediction in the Cauvery Delta Zone, Tamil Nadu, India. Performance is evaluated using metrics such as R^2 , RMSE, MAE, MSE, MAPE, CV, and NMSE, with the SVM model outperforming others with values of 0.9863, 0.2295, 0.1290, 0.0526, .3439, 0.0255 and 0.0136, respectively. This study also compares SVM's performance with other models in the literature, demonstrating its superior prediction accuracy.

KEYWORDS : Crop yield prediction; Machine Learning Algorithm.

INTRODUCTION:

Agriculture holds a central role in India, covering 60% of the land and providing for 1.2 billion people [1]. To aid farmers, modernizing agriculture is crucial. Crop yield depends on factors like weather, rainfall, water management, and pesticide use. Data mining, machine learning, and deep learning are employed to enhance crop yield and quality. Machine learning learns patterns from data and improves performance by understanding input data consistency and trends. This study uses various machine learning methods—Linear Regression, Random Forest Regressor, and Support Vector Machine Algorithm—for yield prediction across different states and crops.

Machine learning is part of Artificial Intelligence (AI) that learns patterns from data. Models are trained using prior outcomes, and testing is done on unused historical data. ML models can be descriptive

or predictive. Predictive models use past data to anticipate future outcomes, while descriptive models depict present or past situations. Machine learning aids in predicting yields and making informed agricultural decisions. Various algorithms like multivariate regression, decision trees, association rule mining, and artificial neural networks have been used for crop yield prediction [5] [6]. Deep learning, a subset of ML, is gaining traction, using neural networks to predict outcomes from raw data [7].

The study aims to provide data-driven insights for agricultural decisions. The paper structure includes an introduction, methods, dataset description, results, discussion, and conclusion.

Due to population growth and lifestyle changes, food demand is expected to rise by 60% by mid-century [1]. Changing climate affects crop yield, increasing risks for farmers. To address this, accurate forecasting systems

are essential for evaluating crop conditions, types, and yields [2]. Crop yield methods are complex due to various factors influenced by non-linear interactions [3]. Traditional methods based on historical data are accurate but are being complemented by remote sensing techniques and machine learning algorithms [4] [5,6] [7]. Machine learning, including artificial neural networks, decision trees, and regression analysis, is advantageous due to abundant data availability [9]. This study focuses on reviewing existing machine learning algorithms used for yield prediction, with a goal to enhance accuracy.

In the context of yield prediction, agriculture forecasting is vital. Numerous research efforts aim for accurate yield prediction, utilizing various methodologies. Some notable works are detailed below.

LITERATURE REVIEW:

Anticipating the harvest is a supreme task for leaders to navigate quickly in public and congregational situations (such as EU status). An accurate crop forecasting model can help operators decide what to plant and when. There are colorful ways to deal with crop forecasting. This review team examined the conclusions reached about the use of artificial intelligence in crop marking.

When examining returned distributions, one of the rejection criteria is that the distribution is a research or standard review. These prohibited distributions are actually the work of affiliates and will be reviewed in this section. Chlingaryan and Sukarieh played the checksum of nitrogen status assessment by practicing artificial intelligence (Chlingaryan et al., 2018). Paper reasons that prevent progress tracking and ML styles lead to practical organization in livestock. Elavarasan et al. conducted a study on distributions with AI models related to crop forecasting under climate constraints. The work encourages the search for broad boundaries that represent crop yields

(Elavarasan et al., 2018). Liakos et al (2018) shared a review on the application of artificial intelligence in breeding. Data transfer happened by breaking up the board by restoring loot, creature leaders, dip and taint. Li, Lecourt and Minister studied the readiness of organic products to infer the ideal harvest time and yield (Li et al., 2018). Mayuri and Priya were prone to difficulties and approaches suffered in image running and artificial intelligence in breeding and especially in love recognition (Mayuri and Priya, xxxx). Somvanshi and Mishra presented many artificial intelligences and how they work in manufacturing intelligences (Somvanshi and Mishra, 2015). Gandhi and Armstrong shared a review paper on the application of data mining in general navigation management in livestock production. They concluded that longer studies should be completed to understand how to understand data mining related to complex pastoral datasets (Gandhi and Armstrong, 2016a, Gandhi and Armstrong, 2016b). Beulah presented a study on different data mining strategies used in performance forecasting and concluded that performance forecasting can be solved using data mining techniques (Beulah, 2019). Many companies use machine learning (ML) techniques to predict consumer phone usage and to study consumer behavior in supermarkets (Ayodele, 2010). in 2016 (Witten et al.). A few years ago, machine learning began to be used in agriculture (McQueen et al., 1995). One of the most difficult problems in precision agriculture is the prediction of crop production, and several models have been proposed and tested so far. Since many factors such as climate, weather, soil, fertilizer use and seed type affect the yield, this work requires the use of multiple data sets (Xu et al., 2019).

Lismail et al. (2018) [1] created a system to predict a country's emergency preparedness. Climate change is studied using machine learning. Southeast Asia is the focus of research. The processes involved in calculating a predictive index include data collection, data

training, data testing, and index calculation.

The index can be used for forecasting, index validation and index visualization, among other things. The study is conducted as a preventive measure. The regions are notified and the sensitivity index is checked using deep learning.

As in our review of check papers, the critical bones of which are presented in this article, this article is the best SLR to highlight the application of artificial intelligence in crop prediction. The current reviews did not methodically check the labels and their breadth looked at concentrations in a specific part of the crop.

METHODOLOGY:

MACHINE LEARNING ALGORITHM:

Using different data strategies to fore see and manage crop yields, crop yield protection uses machine learning algorithms to reduce risk and enhance agricultural productivity. Machine learning models can offer insights and predictions that aid farmers in making decisions by examining many elements that affect crop growth and health.

A. RANDOM FOREST ALGORITHM:

Random forest is a machine learning technique that is frequently used. It mixes the output of various decision trees to get a single conclusion. Its extensive use is motivated by its flexibility and utility because it can solve classification and regression issues. The next diagram depicts it.

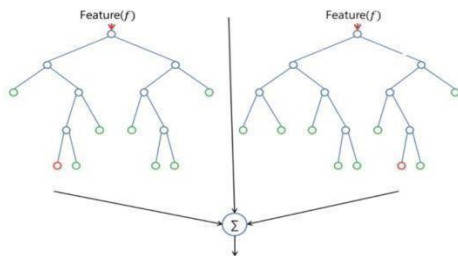


Figure 1: Random Forest Algorithm

The accompanying condition can be utilized to carry out the Arbitrary Backwoods Calculation:

$$RFfi_i = \frac{\sum_{j \in \text{all trees}} \text{norm} fi_{ij}}{T}$$

- In the Arbitrary Timberland model, RFfi sub(i) addresses the significance of elements not entirely settled from all trees.
- The standardized element importance for i in tree j is given by normfi sub(ij).
- T is the general number of trees.

By averaging a few choice trees, Irregular Timberland examples overfit and are less delicate to clamor and exceptions in the information. It offers an element significance metric that can be useful for including choice and information.

B.LINEAR REGRESSION ALGORITHM:

One more class of AI procedures is direct relapse, which especially gains from marked datasets and maps the information focuses to the best-carrying out straight roles. This can be applied to conjecture new datasets. The linear regression is portrayed in the accompanying realistic:

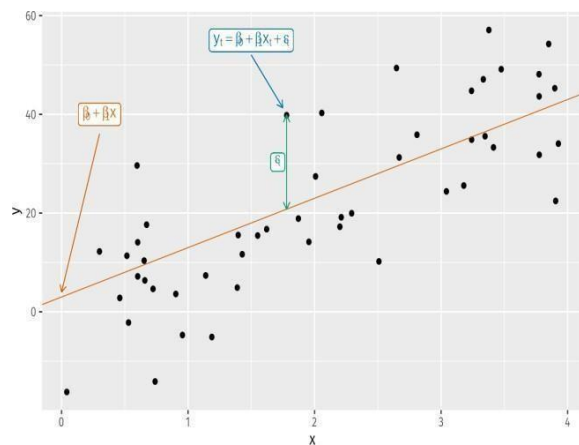


Figure 2: Linear Regression Algorithm

Coming up next is the linear regression condition:

The relapse model takes into consideration the most straightforward direct connection between the figure variable y and a solitary indicator variable x:

$$y_t = \beta_0 + \beta_1 x_t + \epsilon_t$$

The realistic above portrays an imaginary illustration of information from such a model. The capture and incline of the line are demonstrated by the coefficients β_0 and β_1 , separately. The normal worth of y at $x=0$ is addressed by the capture β_0 . The typical extended change in y that outcomes from a one-unit expansion in x is addressed by the slant β_1 .

Observe how the perceptions are scattered around the straight line as opposed to on it. Every perception can be considered at this point, where t is the irregular "blunder" and $\beta_0 + \beta_1 x_t$ is the methodical or made sense of part of the model. The expression "blunder" alludes to a takeoff from the hidden straight-line model instead of a slip-up. It incorporates anything other than x_t that could affect y_t . Since straight relapse simply needs fundamental polynomial math and analytics to be executed, it is a convenient device for exploratory information investigation.

C. SUPPORT VECTOR MACHINE ALGORITHM:

One of the most popular administered learning calculations in AI, the Backing Vector Machine (SVM), is utilized to tackle both order and relapse issues. The SVM calculation's goal is to lay out the best line or choice limit that can partition n- layered space into classes, permitting us to rapidly characterize new data of interest from now on. A hyperplane is the name given to this ideal choice limit.

SVM chooses the outrageous vectors and focuses that guide them in the making of the hyperplane. Support vectors, which are utilized to address these outrageous occasions, structure the reason for the SVM strategy. Consider the outline underneath, where a choice limit or hyperplane is utilized to classify two unmistakable classifications:

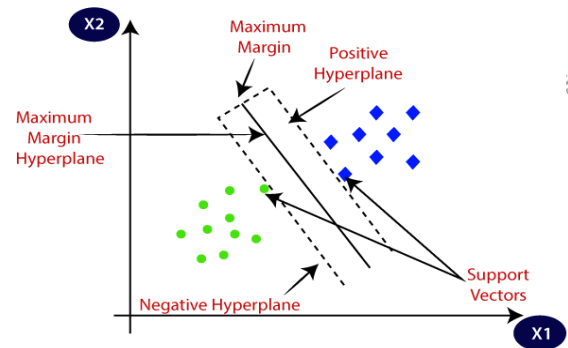


Figure 3: Support Vector Machine Algorithm

The Help Vector Machine (SVM) calculation's equation depends on the idea of recognizing a hyperplane that ideally separates between a few classes of data of interest. The condition for a direct SVM in a double grouping situation is:

$$f(x) = \text{sign}(\sum_{i=1}^N (\alpha_i y_i (x_i, x)) + b)$$

Where:

- The projected class mark for the information point x is addressed by $f(x)$.
- N is help vector.
- α_i are the help vectors' comparing Lagrange multipliers (coefficients).
- y_i is the i th support vector's classname.
- The i th support vector is x_i .
- The spot result of the i th support vector and the information point x is addressed by the maxim " x_i, x ."
- The predisposition term is b.

Tracking down the upsides of α_i and b that expand the edge between the two classes while ensuring that the information focuses are accurately recognized is the point of preparing an SVM.

EXPERIMENTAL ANALYSIS:

In this paper, predication was done by Machine learning algorithms such as Random Forest Algorithm, Linear Regression Algorithm, Support Vector Machine Algorithm. The below graph depicts it as follows:

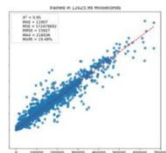


Figure 4: Random Forest

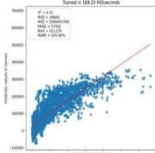


Figure 5: Linear Regression

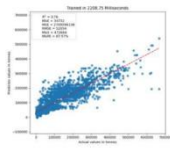


Figure 6: Support Vector Machine

Figure 4: Comparing algorithm technique

This modeling plots display the R^2 , MAE, MSE, RMSE, MAX, and MAPE values. R^2 shows the shortest duration when used to calculate the charts. The regression to the flow figure is indicated by the mean absolute error. In contrast to other regression techniques, the plot in SVM displays projected values and actual values more quickly. Perform one-by-one regression testing using float values and compute R^2 , MAE, MSE, RMSE, MAX, and MAPE to determine crop yields. The cross-validation shows regression testing and reveals improvement in yield prediction. In testing the prediction, linear regression and stochastic gradient descent regression are comparable. Both of the two plots had flawless r^2 and MAPE values throughout testing. In all of the above mentioned figures,

the SVM was assigned in 169.25 milliseconds. The root mean square result was obtained in milliseconds by the R^2 value, which also did the tests.

CONCLUSION:

The prediction of crop yield involves the use of statistical and machine learning techniques. Various machine learning algorithms, such as SVM, SVM, RBFNN, and BPNN, were considered alongside statistical methods like MLR for accurate crop yield prediction. The performance of these methods was evaluated using different metrics.

The key findings are as follows:

- Machine learning algorithms outperformed statistical methods in predicting crop yield accurately.
- Among SVM, RBFNN and BPNN, SVM demonstrated the highest precision in yield prediction.
- The SVM model exhibited strong performance across multiple metrics including R^2 , RMSE, MAE, MSE, MAPE, CV, and NSME, with values of 0.9863, 0.2295, 0.1290, 0.0526, 1.3439, 0.0255, and 0.0136 respectively.
- The runtime of the SVM model was superior at 880 ms compared to other ANN models.
- The proposed SVM model's R^2 metrics showed a 7.53% improvement compared to existing models.
- Tamil Nadu achieved the highest yield of 3191 kg/ha, matching the accuracy of the proposed SVM model's prediction.
- Tamil Nadu's optimal parameters like rainfall, temperature, and pH contribute to its higher yield.
- The recommended SVM algorithm reduces yield risk for paddy cultivation due to its superior performance.

- Future work will explore hybridizing the four machine learning methods and incorporating additional performance metrics.

In summary, the study compared machine learning algorithms and statistical methods for crop yield prediction, highlighting the accuracy and precision of the SVM model. The model's success is attributed to Tamil Nadu's favorable cultivation conditions and it's expected that future research will further enhance these techniques.

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